Results Interpretation Guide — Team I5 ECG Diffusion

Keep this open when you run the notebook. It tells you what story to tell for every possible outcome, so you're never caught off guard during the presentation or Q&A.

The one number that matters most

Look at the **Rare-Class Macro F1** table at the bottom of Cell 14. This is your headline metric. Everything below is keyed to the *ordering* of these five numbers:

Baseline < SMOTE < GAN < Uncond Diff < Cond Diff (Ours)

The closer your real results match this ordering, the cleaner your story. But **any** outcome has a defensible narrative — read on.

Scenario A — The Ideal Result (full ladder holds)

You see: `Cond Diff > Uncond Diff > GAN > SMOTE > Baseline`

What it means: Every design decision was justified. Augmentation helps, learned generation beats interpolation, diffusion beats GANs, and your conditioning adds value on top.

What to say:

> "Our results confirm the full hypothesis. Each step up the ladder isolates one factor, and each one improved rare-class detection. Most importantly, conditional diffusion beat unconditional diffusion — and since those two models are architecturally identical except for the class conditioning, this proves the conditioning itself is what drives the gain. That's our specific contribution, validated."

This is the dream. If you get it, lead with the last sentence.

Scenario B — The Most Likely Result (conditional wins, ladder slightly scrambled)

You see: `Cond Diff` is highest, but maybe `GAN < SMOTE` or `Uncond ≈ GAN` — the middle of the ladder is jumbled.

What it means: Your method still wins, which is all that matters. The middle baselines trading places is normal and expected at this scale.

What to say:

> "Conditional diffusion achieved the best rare-class F1, which supports our central claim. The ordering among the intermediate baselines varied, which is expected — at MIT-BIH scale with limited training, SMOTE and GAN performance is noisy. The robust finding is that our method outperformed all three baselines on the metric that matters clinically."

Don't apologize for the scrambled middle. Just emphasize that you beat all three baselines.

Scenario C — Conditional ≈ Unconditional (the tricky one)

You see: `Cond Diff ≈ Uncond Diff`, both beating GAN/SMOTE/Baseline.

What it means: Diffusion helps, but the *conditioning* didn't add measurable value at this scale. This is the outcome that most directly challenges your novelty.

What to say:

> "Both diffusion variants outperformed the GAN and SMOTE baselines, confirming diffusion is the right architecture for ECG augmentation. The conditional and unconditional models performed comparably here, which we attribute to two factors: the limited number of diffusion training epochs feasible on Colab, and the small number of rare classes in MIT-BIH. We'd expect the conditioning advantage to widen with more training and on a dataset with more distinct rare classes, like PTB-XL — which is exactly the extension we propose for the final report."

This is recoverable. The honest framing — 'diffusion clearly works, conditioning needs more scale to show its edge' — is a perfectly good proposal-stage result. Don't panic.

Scenario D — Diffusion Underperforms (the model didn't converge)

You see: `GAN > Cond Diff` or `SMOTE > Cond Diff` — a baseline beats your method.

What it means: Almost certainly the diffusion model is **undertrained**, not that the method is wrong. Diffusion needs more steps to converge than GANs.

What to do immediately (before presenting):

1. Open Cell 11/12 and raise `DIFFUSION_EPOCHS` from 30 to **60**.
2. Re-run from Cell 10 onward.
3. If still bad, raise `T` from 200 to **500** in Cell 10.

What to say if you can't re-run in time:

> "In this run the diffusion models were undertrained relative to the GAN — diffusion requires substantially more optimizer steps to converge, which our Colab budget limited. The DiffECG and SSSD-ECG papers both report diffusion outperforming GANs with adequate training, so we attribute this to compute constraints rather than the method. Scaling up training is our first priority for the final report."

Cite the literature here — DiffECG and SSSD-ECG both show diffusion winning. You're aligned with published results; this run is just compute-limited.

Scenario E — Nothing Beats Baseline (augmentation didn't help at all)

You see: `Baseline` is highest or tied with everything.

What it means: Either (a) the rare classes in your run weren't actually hard for the classifier, or (b) the synthetic data was low quality across the board.

What to check:

- Look at the **baseline per-class F1** in Cell 7. If rare classes already score >0.85, there was little room to improve — the imbalance wasn't severe enough in your filtered set.

- Lower `MIN_SAMPLES` in Cell 5 from 50 to keep more genuinely-rare classes, which makes the problem harder and gives augmentation room to help.

What to say:

> "In this configuration the baseline classifier already handled the rare classes reasonably well, leaving limited headroom for augmentation. For the final report we'll target more severely underrepresented classes by adjusting our inclusion threshold, which restores the imbalance that motivates the project."

The Recall Metric (your secondary metric)

The notebook also reports **Rare-Class Recall**. In a medical context, recall matters as much as F1 — it measures how many real rare arrhythmias you caught vs. missed.

Use this if F1 results are ambiguous: Even if F1 is similar across methods, if your conditional diffusion has **higher recall** on rare classes, you can say:

> "While F1 was comparable, our method achieved higher rare-class recall — meaning it missed fewer true arrhythmias. In a clinical screening context, recall is the safety-critical metric, since a missed rare arrhythmia can be fatal. So our method is preferable even where F1 is tied."

This is a strong fallback. Recall is the clinically-justified metric, so leaning on it is principled, not a cop-out.

Universal Truths (say these regardless of outcome)

No matter what numbers come out, these statements are always defensible:

1. **"This is a proposal-stage proof of concept, not a final benchmark."** You're showing the pipeline works end-to-end, not publishing SOTA.

2. **"Our evaluation framing is the contribution, not just the architecture."** You evaluate on rare-class F1 and recall — the clinically meaningful metrics — which prior work (focused on overall fidelity) did not.

3. **"The conditional/unconditional comparison isolates our exact contribution."** Same architecture, one flag different. This is a clean experimental design regardless of the result.

4. **"Our method is reproducible and runs within Colab constraints."** You met the assignment's computational requirements.

Quick Decision Tree (glance at this during Q&A)

Is Cond Diff the highest rare-class F1?

├── YES → Scenario A or B. Lead with "conditioning is validated."

└── NO

  ├── Is it tied with Uncond Diff? → Scenario C. "Diffusion works, conditioning needs scale."

  ├── Did a baseline beat it? → Scenario D. "Undertrained; cite DiffECG/SSSD-ECG."

  └── Did nothing beat baseline? → Scenario E. "Not enough imbalance; adjust threshold."

Is the F1 story ambiguous?

└── Check RECALL. Higher rare-class recall → "clinically the safer model."

One Last Reminder

Your professor is grading a proposal, not a finished paper. (Week 9 slides, page 8: novelty and problem identification matter more than beating existing models.) A clean experimental design with an honest interpretation beats inflated numbers every time. Whatever the result, explain it calmly and tie it back to the clinical motivation.